

SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0713

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

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Annexure A

Debugging & Optimisation Task

Code of Conduct:

I Ahamed Kaif S(192321128) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Ahamed Kaif S

Annexure A

Debugging & Optimisation Task

1. A rapidly growing IT company has shifted its operations to a newly constructed sixstorey smart office building. More than **1,200 employees** use wireless laptops, smartphones, IP phones, printers, and IoT devices connected through the company's **Wireless Local Area Network (WLAN)**. The office relies heavily on cloud computing, online meetings, video conferencing, and file sharing, making uninterrupted wireless connectivity essential for daily operations. In addition, the company has a team of field engineers and sales executives who travel across different cities and use **4G/5G Wireless Wide Area Network (WWAN)** services to access the company's servers, customer databases, and business applications from remote locations.

During the last few weeks, employees have reported several connectivity issues. Users experience slow internet speeds during peak working hours, frequent Wi-Fi disconnections while moving between floors, and weak wireless signals in conference rooms. Video conferencing applications often freeze or disconnect, affecting communication with international clients. Network administrators have also observed that some wireless access points are overloaded while others remain underutilized, leading to uneven traffic distribution. Furthermore, field engineers working in rural and highway areas frequently lose WWAN connectivity, delaying the upload of maintenance reports and customer information. Security audits reveal that a few wireless devices are still connected using outdated security protocols, exposing sensitive company information to potential cyber threats. The management has requested the network engineering team to investigate these issues, identify the root causes, and optimize both the WLAN and WWAN infrastructure to ensure high performance, secure communication, and seamless connectivity for all users.

2. A multinational financial organization operates its headquarters, regional offices, and multiple data centers using an **Asynchronous Transfer Mode (ATM) network** to support mission-critical services such as online banking, video conferencing, Voice over IP (VoIP), and real-time financial transactions. The ATM network uses fixed-size **53-byte cells** and establishes **Virtual Circuits (VCs)** to provide reliable and predictable Quality of Service (QoS). Recently, the organization has experienced several performance issues during peak business hours. Customers report delays in online transactions, voice calls suffer from jitter and occasional packet loss, and video conferencing sessions experience reduced quality. Network monitoring indicates that several ATM switches are operating at high utilization, some Virtual Circuits are congested, and available bandwidth is not being efficiently allocated among different traffic classes. Engineers also discover that certain applications are using inappropriate ATM service categories, leading to increased delay and inefficient resource utilization. The IT department has assigned you as the network engineer to investigate the ATM network, identify the faults, debug the performance issues, and recommend optimization strategies to restore efficient, reliable, and high-quality communication across the enterprise.
3. A multinational e-commerce company has interconnected its headquarters, regional offices, warehouses, and cloud data centers using a **Multiprotocol Label Switching (MPLS) network** to ensure fast, secure, and reliable communication. The MPLS infrastructure supports critical business applications, including online order processing, enterprise resource planning (ERP), video conferencing, Voice over IP (VoIP), cloud-based inventory management, and real-time customer support. Recently, the organization has experienced frequent performance issues

during peak business hours. Employees report slow access to cloud applications, delayed order processing, poor-quality voice calls, and interruptions during video conferences. Network monitoring reveals that some Label Switched Paths (LSPs) are heavily congested, while other paths remain underutilized. The IT team also discovers that several routers have incorrect label mappings, causing inefficient routing and increased latency. In addition, traffic engineering has not been properly configured, resulting in uneven bandwidth utilization across the MPLS backbone. Since these issues are affecting customer satisfaction and business operations, the network administrator has asked you to investigate the MPLS network, identify the faults, debug the performance problems, and recommend suitable optimization techniques to improve the overall efficiency and reliability of the network.

4. A multinational logistics company operates an automated warehouse where barcode scanners, RFID readers, autonomous robots, and inventory management systems continuously exchange data over a computer network. Every day, thousands of shipment records, customer details, and inventory updates are transmitted between warehouse terminals and the central server. Recently, the company has observed that some shipment records are arriving with incorrect information, barcode scans occasionally fail, and inventory quantities in the database do not always match the physical stock. During peak hours, network monitoring reports an increase in transmission errors caused by electrical interference, signal attenuation, and network congestion. Although some corrupted frames are detected and retransmitted, others remain undetected, resulting in incorrect database entries and delayed order processing. The IT department suspects that the current error detection and correction mechanisms are not configured efficiently. As the network engineer, you are assigned to investigate the transmission errors, identify the weaknesses in the existing error control techniques, and recommend suitable debugging and optimization strategies to improve the reliability and accuracy of data communication.
5. A leading online education platform conducts live classes and online examinations for more than **15,000 students** across different locations. During examinations, the central server simultaneously sends question papers, images, videos, and answer submission acknowledgements to all students over the network. The server is equipped with a high-speed **10 Gbps** connection, whereas many students access the system using slower broadband or mobile internet connections. During peak hours, students experience incomplete downloads, delayed page loading, duplicate packets, and repeated retransmissions. Some students receive the next set of data before their devices finish processing the previous one, causing receiver buffer overflow and application crashes. Network administrators also observe an excessive number of retransmissions, increased latency, reduced throughput, and poor bandwidth utilization. The current flow control mechanism is not effectively regulating the transmission rate between the sender and receivers. As the network engineer, you have been assigned to investigate the communication problems, identify the causes of inefficient flow control, debug the existing system, and recommend suitable optimization techniques to ensure reliable and efficient data transmission during high-traffic periods.

ANSWERS:-

1. WLAN & WWAN Network Issues

Problems

- Slow internet during peak hours.
- Frequent Wi-Fi disconnections between floors.
- Weak signal in conference rooms.
- Video conferencing freezes.
- Uneven Access Point (AP) load.
- WWAN connectivity loss in remote areas.
- Outdated wireless security protocols.

Root Causes

- Overloaded wireless access points.
- Poor AP placement and channel interference.
- Insufficient wireless coverage.
- Lack of roaming optimization.
- Weak cellular signal in rural areas.
- Use of outdated security (WEP/WPA).

Debugging Steps

- Perform wireless site survey.
- Monitor AP utilization and client count.
- Analyze signal strength (RSSI/SNR).
- Check roaming logs.
- Test WWAN signal quality.
- Audit wireless security settings.

Optimization Techniques

- Add or relocate access points.
- Configure load balancing.
- Enable fast roaming (802.11r/k/v).
- Optimize channel allocation.
- Upgrade to WPA3 security.
- Use signal boosters or dual-SIM WWAN devices.

2. ATM Network Performance Issues

Problems

- Delayed online transactions.
- VoIP jitter and packet loss.
- Poor video conferencing quality.
- Congested Virtual Circuits.
- Uneven bandwidth allocation.

Root Causes

- High ATM switch utilization.
- Congested Virtual Circuits.
- Incorrect ATM service categories.
- Poor QoS configuration.
- Inefficient bandwidth allocation.

Debugging Steps

- Monitor ATM switch statistics.
- Check VC utilization.
- Analyze QoS parameters.
- Verify service category configuration.
- Identify congested links.

Optimization Techniques

- Reconfigure Virtual Circuits.
- Allocate bandwidth based on QoS.
- Use correct ATM service categories (CBR, VBR, ABR, UBR).
- Balance traffic across switches.
- Upgrade switch capacity if needed.

3. MPLS Network Issues

Problems

- Slow cloud application access.
- Delayed order processing.

- Poor VoIP quality.
- Video conference interruptions.
- Congested Label Switched Paths (LSPs).

Root Causes

- Incorrect label mappings.
- Poor Traffic Engineering.
- Uneven bandwidth utilization.
- Congested MPLS paths.
- Router configuration errors.

Debugging Steps

- Verify MPLS label tables.
- Check LSP utilization.
- Monitor router CPU and interfaces.
- Validate Traffic Engineering configuration.
- Analyze latency between sites.

Optimization Techniques

- Correct MPLS label mappings.
- Configure MPLS Traffic Engineering (TE).
- Load balance traffic across LSPs.
- Prioritize critical applications using QoS.
- Optimize routing policies.

4. Error Detection & Correction

Problems

- Corrupted shipment records.
- Barcode scan failures.
- Inventory mismatch.
- Transmission errors.
- Data corruption.

Root Causes

- Electrical interference.
- Signal attenuation.

- Network congestion.
- Weak error detection mechanisms.
- Inefficient retransmission.

Debugging Steps

- Monitor frame error rate.
- Capture corrupted packets.
- Check CRC errors.
- Analyze retransmission logs.
- Test communication channels.

Optimization Techniques

- Implement stronger CRC.
- Use Forward Error Correction (FEC).
- Improve Automatic Repeat Request (ARQ).
- Replace faulty cables/devices.
- Reduce interference using shielded cables.

5. Flow Control Issues

Problems

- Incomplete downloads.
- Receiver buffer overflow.
- Duplicate packets.
- High retransmissions.
- Increased latency.
- Poor throughput.

Root Causes

- Sender transmits faster than receiver.
- Inefficient flow control protocol.
- Small receiver buffers.
- Heavy network congestion.
- Poor bandwidth utilization.

Debugging Steps

- Monitor buffer utilization.

- Analyze TCP window size.
- Check retransmission statistics.
- Measure throughput and latency.
- Capture packet flow using Wireshark.

Optimization Techniques

- Implement Sliding Window protocol.
- Use Dynamic Window Scaling.
- Apply congestion control algorithms.
- Increase receiver buffer size.
- Optimize bandwidth allocation.
- Enable selective acknowledgements (SACK).

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained

Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, wellstructured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation